Assessing the feasibility of creating realistic type Ia supernova simulations in state-of-the-art cosmological simulations

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1 Abstract

Obtaining credible predictions of supernova type Ia (SNIa) rates from simulated galaxy properties enables accurate studies of SNIa progenitors and the impact of the large-scale environment on cosmological parameters inferred from SNeIa. It has been shown that different modelling prescriptions for galaxy evolution can have a significant impact on the derived transient rates. Here, we derive the SNIa rates of individual galaxies at low redshifts assuming the empirical Childress+14 star-formation history and semi-analytic star-formation histories from the state-of-the art `MultiDark` (`MDPL2` `SAG`, `SAGE` and `GALACTICUS`) simulations.

2 Introduction

Accurate estimates of simulated SNIa events are crucial for a wide range of scientific applications. On the astrophysical side, population synthesis codes can be validated by comparing their predictions with observed rates [16]. Additionally, delay-time distributions (DTDs) can be constrained by examining the correlation between SNIa rates and the properties of their host galaxies [2]. On the cosmological side, correlations of simulated SNIa events with the large-scale structure can probe the effect of the local environment on cosmological parameters inferred from SNeIa [3, 18], or shed light on the clustering properties of SNIa hosts [19].

Realistic SNIa rate simulations require adequate modelling of baryonic physics, which is either provided by more expensive hydrodynamical simulations [12, 11], or less expensive semi-analytic models (SAMs) which are applied to simulations with gravity only and then evolve galaxies in dark matter haloes. Such semi-analytic recipes are `SAG` [15], `SAGE` [5], `GALACTICUS` [1] or `GALFORM` [9]. Due to their relatively inexpensive nature, SAMs have been widely adopted to approximate galaxy formation and evolution in large cosmological volumes in state-of-the-art simulation suites, like `Uchuu` [7], `MultiDark` [8] or `Millenium` [14].

In this work, we derive SNIa rates from different combinations of the galaxy evolution prescriptions `SAG`, `SAGE` and `GALACTICUS` to understand the feasibility of launching a research programme aimed at the creation of state-of-the-art SNIa simulations in the cosmic large-scale structure. To this end, we compare the rates as a function of galaxy properties to observations in the low-redshift Universe. In Section 3, we describe the galaxy evolution models compared and our choices on the SNIa rate model and event occurrence. In Section 4, we present our results on the four modelling variants we assume. In Section 5, we conclude with our results and future outlook.

3 Method

In our analysis we employ four galaxy evolution prescriptions, which we describe below.

- Childress+14: In the nearby Universe, star-formation rates (SFRs) strongly correlate with stellar mass, while their relationship exhibits minimal dependence on redshift. Gas which is converted to stars in a galaxy will be returned to the interstellar medium at a high rate initially and will then cease. Ultimately, galaxies lose their ability to create stars and exceed a redshift dependant quenching threshold. A formula constructed by hand to model the dependence of the SFR on stellar mass satisfying the above physical process was proposed by [4]

$$\Psi(M_*, z) = \left(\frac{M_*}{10^{10}}\right)^{0.7} \frac{\exp\big(1.9z\big)}{\exp\left[1.7(z-2)\right] + \exp\left[0.2(z-2)\right]} \quad \left[M_\odot \mathrm{yr}^{-1}\right], \tag{1}$$

ensuring the SFR scales appropriately with both stellar mass and redshift. After multiplying by a quenching penalty P_Q,

$$p_Q(M, z) = \frac{1}{2}\left[1 - \mathrm{erf}\left(\frac{\log(M) - \log\left(M_Q(z)\right)}{\sigma_Q}\right)\right], \tag{2}$$

where $\log\big(\frac{M_Q(z)}{M_\odot}\big) = 10.077 + 0.636z$, and accounting for the mass lost in each time step ΔM_*, a final prescription for SFR as a function of redshift and stellar mass can be written as:

$$\frac{M_*(t+\Delta t) - M_*(t)}{\Delta t} = p_Q\left(M_*(t), z(t)\right)\, \Psi\left(M_*(t), z(t)\right) - \frac{\Delta M_*}{\Delta t}. \tag{3}$$

The mass loss ΔM_* is given by,

$$\Delta M_* = \int_0^{t_f} \Psi(M_*(t-\tau), z(t-\tau)) \cdot \left(f_{\mathrm{ml}}(\tau + \Delta t) - f_{\mathrm{ml}}(\tau)\right)\, d\tau, \tag{4}$$

where $f_{\mathrm{ml}} = 0.046 \ln\left(\frac{\tau}{0.276\,\mathrm{Myr}} + 1\right)$.

- `MDPL2` `SAG`, `SAGE`, `GALACTICUS`: These three models were selected for application to the `MDPL2` catalogues for their ability to reasonably model gas / metallicity properties, reproduce multiple observables and capture the evolution of SFRs, respectively. We refer the reader to [8] for a technical comparison of these models. Here, we focus on the impact their technical differences have on the predicted galaxy stellar masses and star-formation histories, which may have a significant impact on the SNIa rates. The `MDPL2` simulation itself covers a comoving volume of 1 h-1 Gpc and adopts a planck based cosmology. In our analysis we concentrate on galaxies with redshifts $0 \leq z \leq 0.15$, and $M_* > 10^7\, M_\odot/h$. Despite sharing the same underlying N-body simulation, the three SAMs differ in how they treat key processes: `SAG` includes an elaborate chemical-enrichment scheme, `SAGE` emphasizes broad reproducibility of observed galaxy properties, and `GALACTICUS` offers flexible treatments of disk/bulge formation and feedback. Consequently, the predicted stellar masses and SFHs can differ among these models.

3.1 Type Ia supernova rate models

- A+B model [20]:

  The development of this model followed the finding that SNeIa can arise both from old ('tardy') and young ('prompt') progenitors. In its simplest form the model gives SNIa rates R_{IA} as:

  $$R_{IA} = A \times M_* + B \times \mathrm{SFR}, \tag{5}$$

  where the older stellar populations and longer time delay SNeIa are given by the $A \times M_*$ section of the rate of the calculation, and likewise the young progenitors which explode on shorter timescales are accounted for by the $B \times \mathrm{SFR}$ term of the equation. For this model we take both M_* and SFR at $z = 0$; an approximation we expect to be justifiable since at low redshifts galaxy SFR is changing slowly on timescales relevant to prompt SNeIa. Similarly using M_* will still be a suitable measure for the older population indicating that even under these assumptions we will be able to yield reasonable first order rate estimates.

- integrated SFH model [21]:

  DES paper: the A+B approximation arises as a direct consequence of the combination of a power-law DTD with the average SFHs of galaxies [10], assuming that the shape of the DTD is reasonably close to a τ^{-1} function, extending from prompt times ($\sim 10^8$ yr) out to several Gyr. The prompt component is proportional to the amount of on-going star-formation at the epoch of observation, and the tardy component is caused by the fact that massive galaxies experienced high SFRs several Gyrs ago. Recent advances in integral field spectroscopy allow for a more granular look at

star formation within galaxies, allowing us to recreate spatially resolved spectra that can be used to reconstruct SFHs within host galaxies. With SFHs for each galaxy we are able to implement an integral model in order to calculate SNIa rates, with the following relationship:

$$R_{IA} = \int_{t_0}^{t_f} \psi\big(t_0 - \tau\big)\, \Phi(\tau)\, d\tau \tag{6}$$

where ψ is the SFH evaluated at a time τ before the epoch at which the galaxy is observed t_0, and $\Phi(\tau)$ is the delay time distribution as described by the function:

$$\Phi(\tau) = A \left(\frac{\tau}{\mathrm{Gyr}} \right)^{\beta}, \tag{7}$$

for all $\tau \geq t_p$ where τ is the time since a burst of star formation and t_p is the prompt time.

3.2 Event sampling from derived rates

Before comparing our calculated rates to observations in the nearby Universe, we first approximate the occurrence of SNeIa in the large-scale structure by modelling their distribution as a Poisson process. This assumption is motivated by the fact that SNeIa are rare events, and is is supported by the statistical treatment in [6]. Specifically, the calculation of the volumetric SN rate which inherently treats each SN event as an independent random occurrence, as well as Monte Carlo simulations which placing millions of SNe Ia at random locations in time and space, align with a Poisson process assumption. To obtain a discrete count of SNeIa per galaxy, from our list of SNIa rates, we use a Poisson sampling approach, where the number of SNeIa occurring in each galaxy over a time interval ΔT follows:

$$P(k) = \frac{\lambda^k e^{-\lambda}}{k!} \tag{8}$$

Specifically, we take a snapshot interval $\Delta T = 1$ yr at $z \approx 0$ and for each galaxy compute the expected number of events $\lambda_i = \Delta T * R_{IA,i}$, where $R_{IA,i}$ is the calculated SNIa rate for each galaxy. We apply this approach for all galaxies in our sample with $M_* > 10^7\, M_\odot/h$. This mass cut is justified by observations indicating that nearly all SNIa hosts have stellar masses above this threshold [17]. Applying this constraint ensures that we include galaxies that are observed to be relevant as SNIa hosts while minimizing contamination from galaxies with negligible SNIa rates. This process gives a set of galaxies containing > 0 SNIa events (SNIa hosts), along with their relative stellar masses and star formation rates.

4 Results

In the following section we present results for the `SAG`, `SAGE` and `GALACTICUS` simulations. Each graph shows results in a format as taken from Tsaprazi and Heavens (2025, Figure 3.) [18].

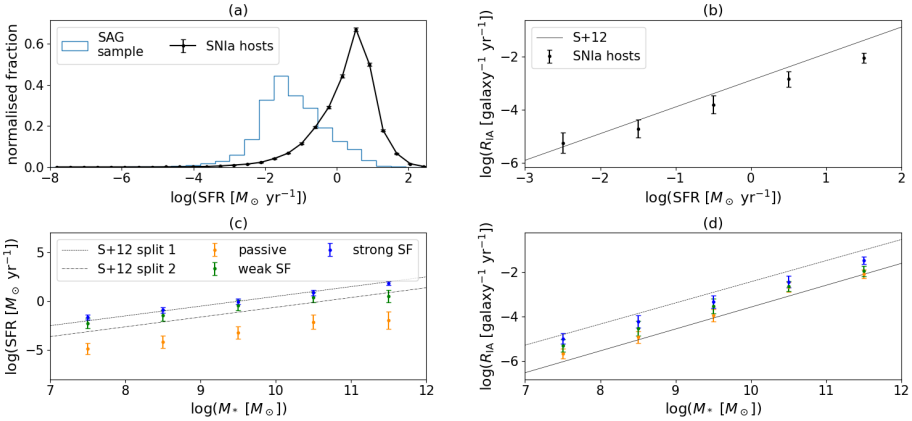


Figure 1: (a) Distributions of the `SAG` SNIa host star-formation rates. In light blue, we show the SFR distribution of the entire `SAG` galaxy sample. (b) SNIa rate per galaxy as a function of the star-formation rate. The continuous line is from Smith et al. (2012, Figure 4)[13]. (c) SNIa hosts on the star-formation rate - stellar mass plane, coloured according to their specific star-formation rate. The black dashed line separates highly from moderately star forming galaxies in Smith et al. (2012, Figure 2). The dashed dotted one encloses all weakly star-forming hosts in that plot. (d) SNIa rate per galaxy as a function of host stellar mass. The lines are from Smith et al. (2012, Figure 3). The error bars indicate 1σ uncertainty over the hosts in 300 Poisson realisations.

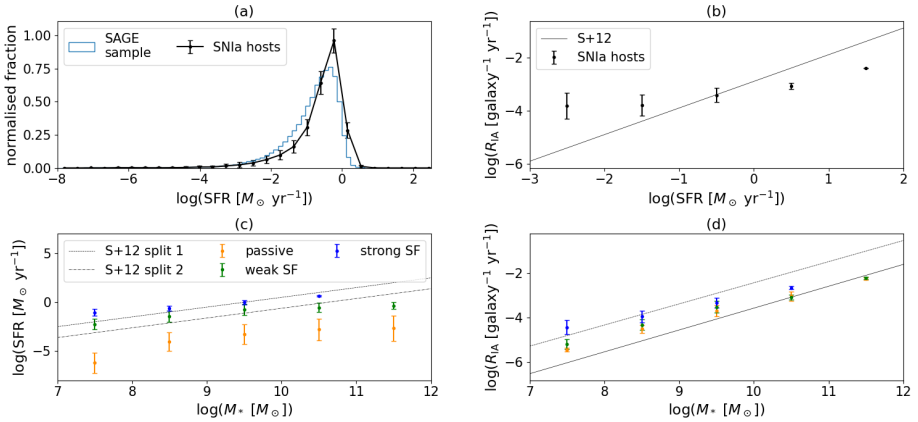


Figure 2: (a), (b), (c) and (d) as above in Figure 1, with analysis done for a population of `SAGE` SNIa hosts.

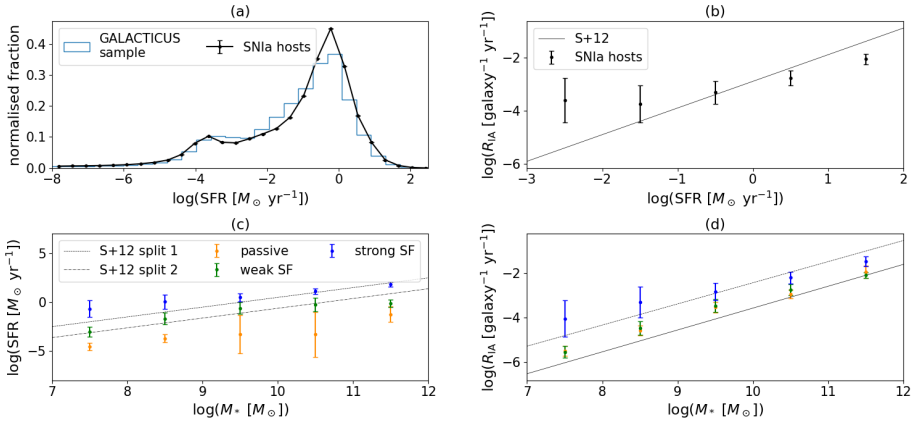


Figure 3: (a), (b), (c) and (d) as above in Figure 1, with analysis done for a population of `GALACTICUS` SNIa hosts.

4.1 Comparison of MDPL2 results

Comparing our results between MDPL2 datasets for each subplot, we can comment on how reasonable the SNIa rates seem for host galaxies as a function of stellar mass and SFR.

4.1.1 (a)

The largest variation between the distribution of the hosts and the overall sample seems to be for `SAG` in Figure 1. This suggests that the SNe Ia host galaxies

selected by our model do not follow the same mass and star formation rate distribution as the full galaxy population. Promisingly, we observe a bimodal distribution in Figure 3 for `GALACTICUS`, which aligns with our expectations for distinct SNIa progenitor channels [20], indicating greater reliability in the rates produced from our analysis of `GALACTICUS`.

4.1.2 (b)

Out of all of our data, we see the largest deviation from our expectations in subplot (b). While results from `SAG` line up very well visually with the line from Smith et al. (to within some constant), we observe a significant deviation from the line at smaller star-formation rates in Figure 2 and Figure 3 (`SAGE` and `GALACTICUS`). Deeper analysis into the distribution of SNIa rates within the lower SFR bins would be needed to make comment on whether these deviations might arise from the fact that our sample of galaxies from the dataset isn't representative of observed SNIa hosts.

4.1.3 (c) and (d)

Subplots (c) and (d) align very well visually with the lines obtained from Smith et al (2012, Figures 2/3) [13]. Passive, weakly and strongly star forming all lie within some small error on the correct side of the these splits by specific star-formation rate, indicating some realiability in all three of the MDPL2 datasets when it comes to simulating supernova formation rates.

We attach a link to the github repository containing the code used for our data analysis. [1]

5 Conclusions

Our goal was to obtain credible SNIa rates from various simulated datasets. Applying an A+B model as described in Section 3.1, and using a Poisson sampling approach to derive SNIa hosts from our list of rates, we were able to obtain a set of host galaxies along with their respective SNIa rates, stellar masses and star formation rates. Comparison to relevant observations from the Sloan Digital Sky Supernova Survey-II [13], yielded reasonable fits for hosts from all three of `SAG`, `SAGE` and `GALACTICUS`. The apparent consistency between real observations and our simulated Type Ia supernova properties, demonstrated a reliability in Type Ia supernova simulations in cosmological simulations. In conclusion, this preliminary investigation showed us that it is possible to create supernova simulations at the individual galaxy basis using state-of-the-art simulations of the cosmic large-scale structure - a next step in building an interface between supernova science and the large-scale structure of the universe.

[1] github repository `https://github.com/francesco-calice/UROP-Supernovae`

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